

WordEmbeddingsSearch

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@version Java 21

Description

This is a console-based application that creates Arrays from a user specified text file and outputs into a user specified text file the top ten similar words to a user specified word, using vector comparison algorithms to search and return them,

To Run

Using an IDE navigate to the relevant folder the application files are saved in and open the Runner.java file.

Click on run and in the terminal section of the Run window a menu will appear. Follow the on-screen instructions based on the menu options and enter the selections in the console window.

Features

- A clear and structured menu interface for users to interact with is provided.
- Specify input and output text files.
 - Input the file path for the relevant embedding file.
 - Input the file path for the destination for the results file.
 - Removal of any additional characters that can occur when copy and pasting file paths.
- Input validation is handled to ensure the necessary fields are filled before proceeding with the search.
- A vector comparison search is ran using specific algorithms comparing a user inputted word with the contents of the user inputted embeddings file.
- The results file is checked for data and if any data is contained it, it is deleted before outputting the new results to it.
- There are safeguards in place to handle errors that may occur.
 - Validation the user inputted word is on the list.
 - Validation that selected option is a menu option.
 - Validation of file paths and validation of file creation.
 - Validation Arrays are same length.
 - Validation calculations return values ≥ 0 .
 - The user input word is not included as part of the result.
 - Validation results file can be deleted and written to.